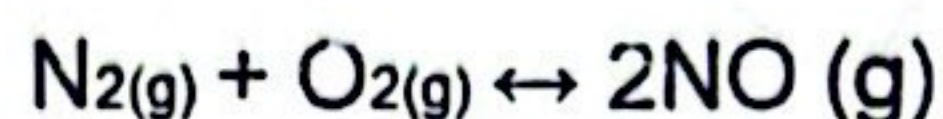


KNOWLEDGE	INQUIRY	COMMUNICATION	APPLICATION
/ 10	11 / 12	8 / 8	8 / 8

INQUIRY – Record your answer according to the correct significant digits, as per this course. If an assumption is made, show the 100 Rule. (12 marks)

1. Consider the following reaction:

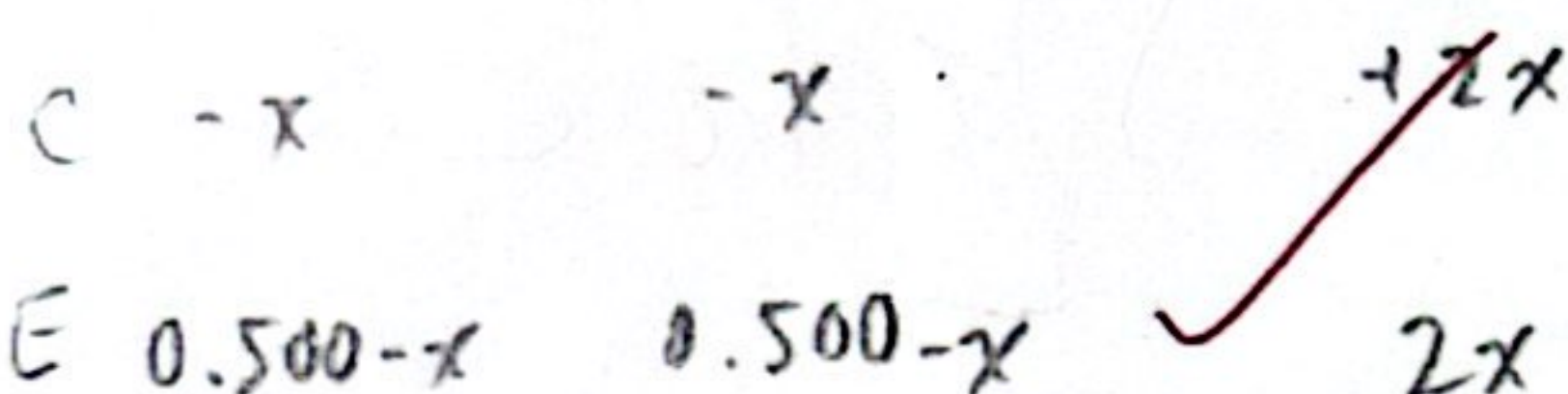
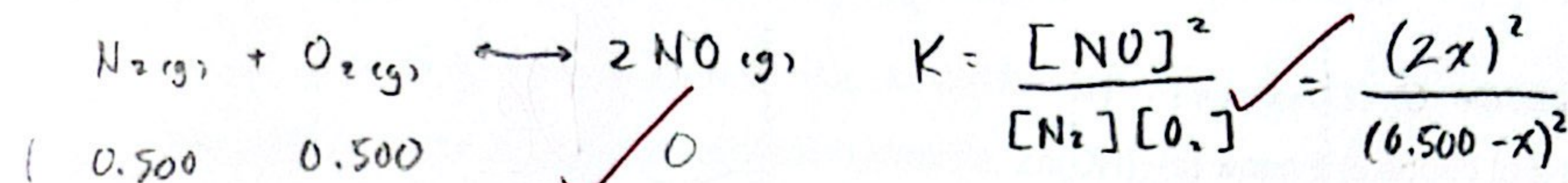


$$K = 0.088 \text{ at } 3727^\circ\text{C}$$

A reaction mixture at 3727°C initially contains 1.00 mol of nitrogen gas and 1.00 mol oxygen gas in a 2.00 L flask.

Find the concentrations of all reactants and products at equilibrium at this temperature. (6 marks)

$$C = \frac{n}{V} \quad C_0 = \frac{1.00 \text{ mol}}{2.00 \text{ L}} = 0.500 \text{ mol/L} \quad C_N = \frac{1.00 \text{ mol}}{2.00 \text{ L}} = 0.500 \text{ mol/L}$$



$$K = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} = \frac{(2x)^2}{(0.500-x)^2}$$

$$\sqrt{0.088} = \sqrt{\frac{(2x)^2}{(0.500-x)^2}}$$

$$0.296648 = \frac{2x}{0.500-x}$$

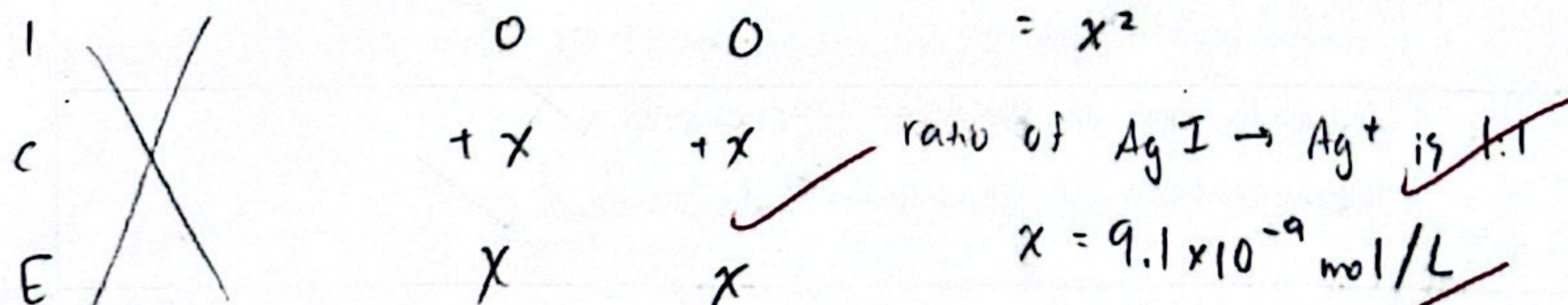
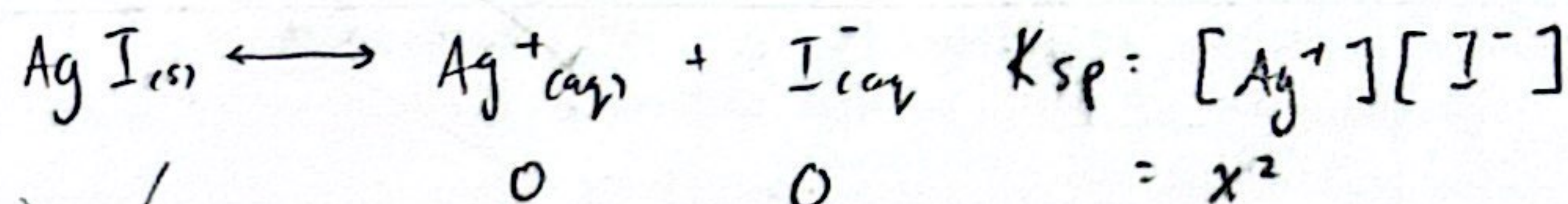
$$0.14832 - 0.296648x = 2x$$

$$0.14832 = 2.296648x$$

$$0.06458 = x$$

$$[\text{NO}]_e = 2(0.06458) = 0.129 \text{ mol/L} \quad [\text{N}_2]_e \text{ and } [\text{O}_2]_e = 0.500 - 0.06458 = 0.435 \text{ mol/L}$$

2. The solubility of silver iodide is $9.1 \times 10^{-9} \text{ mol/L}$. What is the solubility product constant (K_{sp}) at 25°C ? (3 marks)



$$x = 9.1 \times 10^{-9} \text{ mol/L}$$

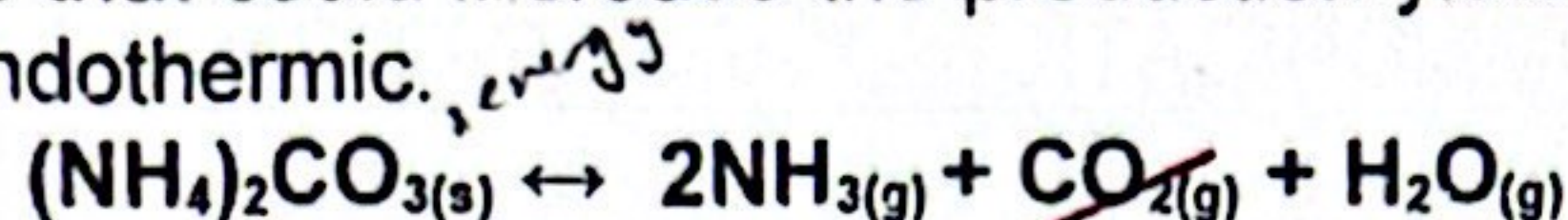
$$K_{sp} = (9.1 \times 10^{-9})^2 = 8.3 \times 10^{-17}$$

3. For each example, complete the chart and predict the direction the equilibrium is shifted by the change imposed (L = left or R = right). If there is not shift, write NO SHIFT: (3 marks)

EQUATION	CHANGE IMPOSED	DIRECTION OF SHIFT
a) $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \leftrightarrow 2\text{CO}_2(\text{g})$	Volume is increased	left ✓
b) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \leftrightarrow 2\text{NH}_3(\text{g}) \quad \Delta H = -92 \text{ kJ}$	Temperature is decreased	right ✓
c) $\text{PO}_4^{3-}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \leftrightarrow \text{HPO}_4^{2-}(\text{aq}) + \text{OH}^-(\text{aq})$	HCl(aq) is added	no shift ✓

COMMUNICATION (8 marks)

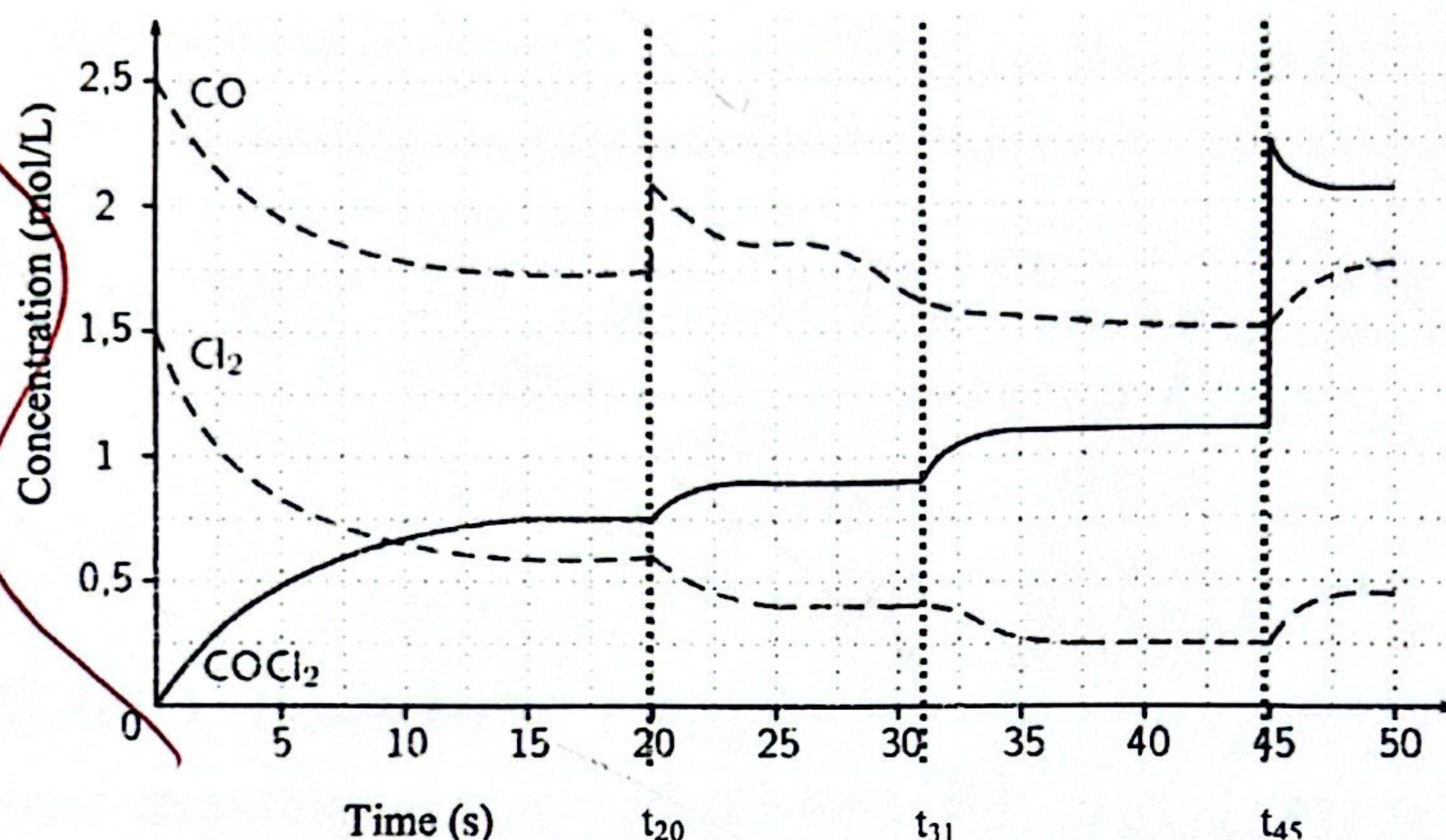
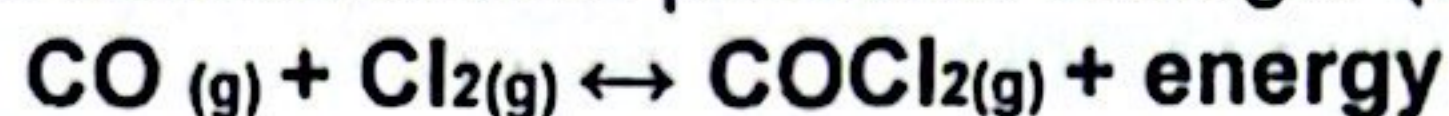
1. Suggest 2 different factors that could increase the production yield of ammonia gas in the reaction below. (2 marks). This reaction is endothermic.



Decrease the concentration of $CO_{2(g)}$ or $H_2O_{(g)}$ (or remove $NH_{3(g)}$ from the system)

Increase the temperature of the system

2. Identify the nature of the change imposed on the equilibrium system, shown in the graph below, at each of the times indicated t_{20} , t_{31} , t_{45} . Also include the direction the equilibrium would shift to account for this change and state how the concentration of reactants and products change. (6 marks)



	Shift	Concentration Changes of reactants and products
t_{20}	right - increased concentration of $CO_{(g)}$	Concentrations of products increased Concentration of Cl_2 would decrease (reactant) After initial jump, concentration of CO also decreases (reactant)
t_{31}	right - decreased temperature of system	Concentration of product increased Concentrations of reactants decreased
t_{45}	left - increased concentration of $COCl_{2(g)}$	After initial spike, concentration of $COCl_2$ decreases (product) Concentrations of reactants increased

APPLICATION – Record your answer according to the correct significant digits, as per this course. If an assumption is made, show the 100 Rule. Please answer in point form where appropriate. (8 marks)

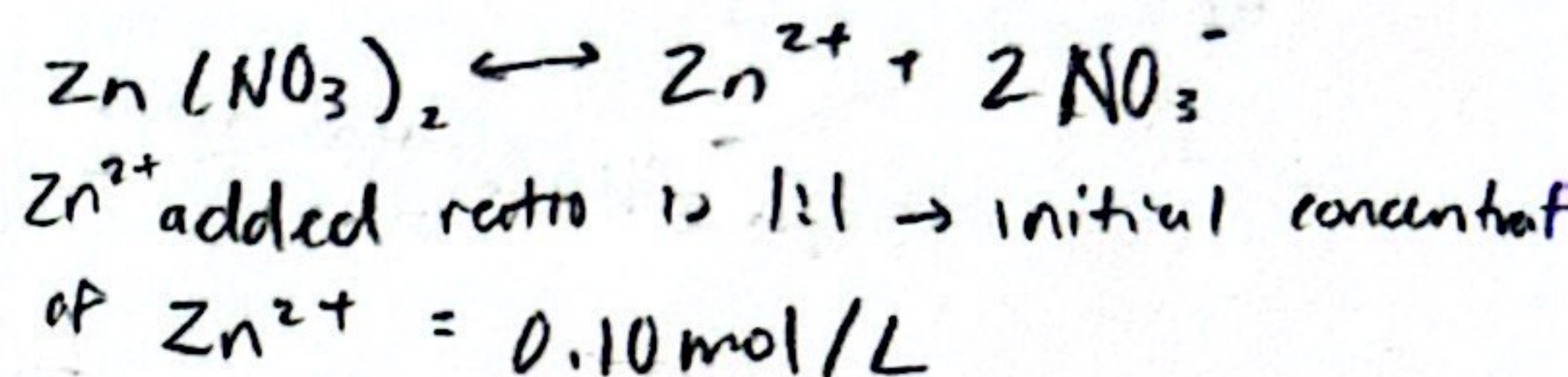
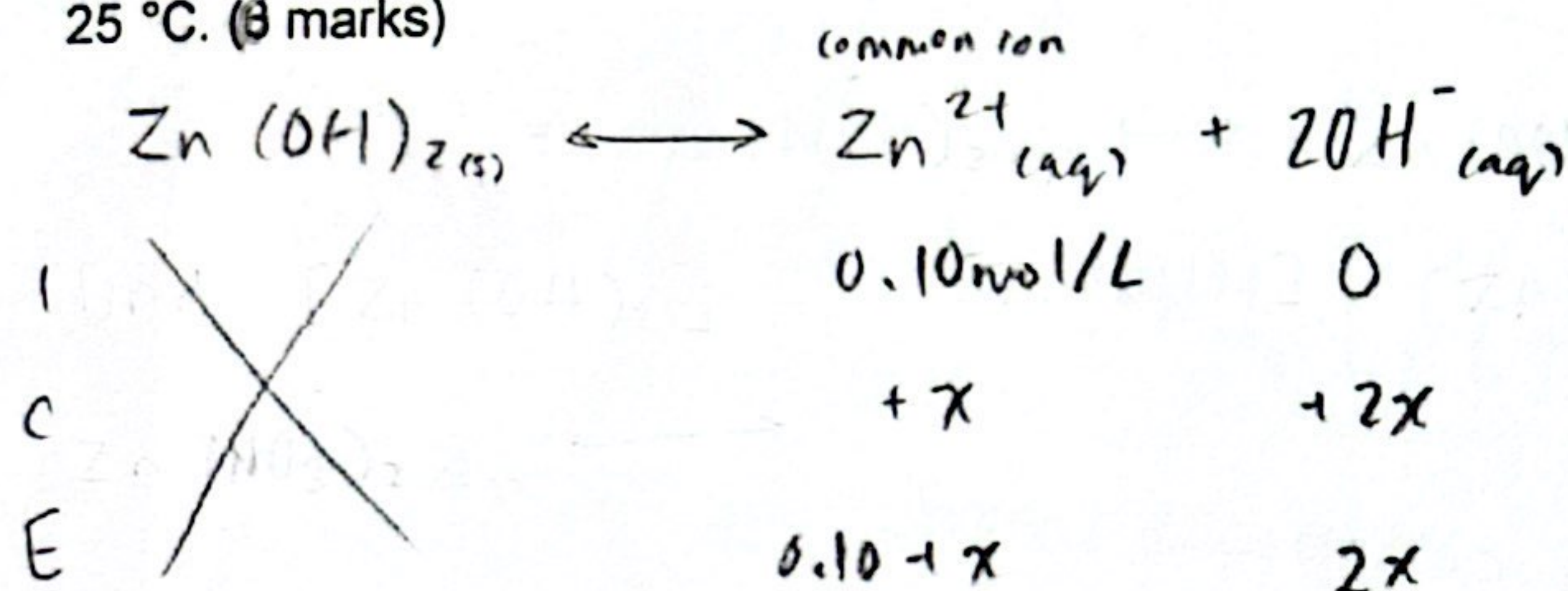
1. Hydrogen gas, $H_2(g)$, has several advantages and disadvantages as a potential fuel. Hydrogen can be obtained by the thermal decomposition of water at high temperatures:



Evaluate the practicality of thermal decomposition of water to obtain hydrogen gas. (2)

- Since K_{eq} is incredibly small, the reverse equation is favoured ✓
- reactants are favoured → less products → less H_2 produced
- since $K_{eq} = \frac{[products]}{[reactants]}$ and K_{eq} is so small, that means the concentration of reactants at equilibrium is far greater than that of products
- means REACTANTS are favoured, equilibrium is positioned on the left
- ∴ Since the reaction favours the creation of reactants so much, barely any H_2 is made so it is impractical (product)

2. Calculate the molar solubility of solid zinc hydroxide, $Zn(OH)_2(s)$ when it is added to a 0.10 mol/L solution of zinc nitrate at $25^\circ C$. (8 marks)



K_{sp} of $Zn(OH)_2$ (from table) = 7.7×10^{-17}

initial concentration of Zn^{2+} / K_{sp}

$$= 0.10 \text{ mol/L} / 7.7 \times 10^{-17}$$

$$= 1.3 \text{ mol/L} > 100$$

∴ 100 rule applies.

$$[Zn^{2+}] = 0.10 + x$$

$$= 0.1000000139 \text{ mol/L}$$

$$K_{sp} = [Zn^{2+}][OH^-]^2$$

$$= (0.10 + x)(2x)^2$$

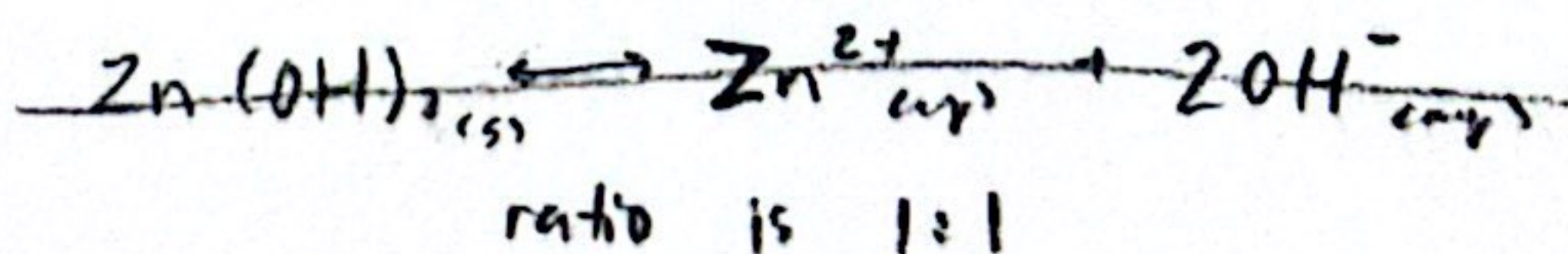
$$= (0.10)(4x^2)$$

$$7.7 \times 10^{-17} = 0.40 x^2$$

$$\sqrt{1.925 \times 10^{-16}} = x^2$$

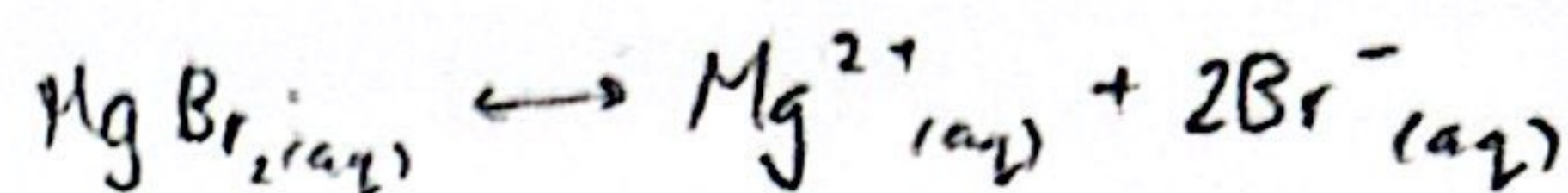
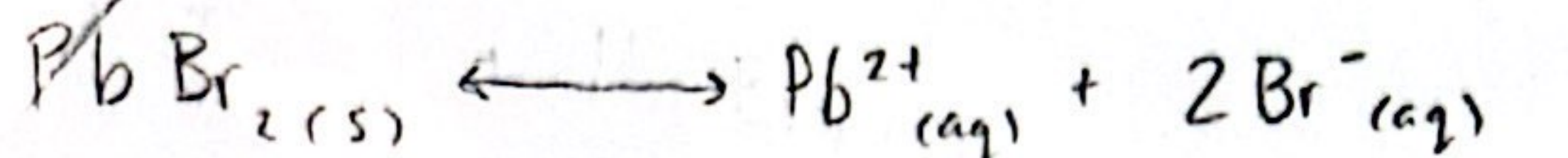
$$1.387 \times 10^{-8} = x$$

$$1.4 \times 10^{-8} = [Zn(OH)_2]$$



Can't use the 0.10x because it doesn't tell us anything about the concentration of $Zn(OH)_2$ itself, the 0.1 is from Zn^{2+} only

2. Calculate the molar solubility of solid lead (II) bromide when it is added to a 0.10 mol/L solution of magnesium bromide at 25 °C. (3 marks)



Ratio of MgBr_2 to Br^- is 1:2

concentration of Br^- ions is $0.10 \times 2 = 0.20 \text{ mol/L}$

	0	0.20 mol/L
	+ x	+ 2x
	x	0.20 + 2x

$$K_{sp} = [\text{Pb}^{2+}][\text{Br}^-]^2$$

From reference table: K_{sp} of $\text{PbBr}_2 = 6.6 \times 10^{-6}$

$$K_{sp} = (x)(0.20 + 2x)^2$$

Initial concentration of Br^- / K_{sp} : (check 100 rule)

$$0.20 \text{ mol/L} / 6.6 \times 10^{-6} = 30303.03 > 100$$

$$6.6 \times 10^{-6} = x(0.20)^2$$

100 rule applies

$$\frac{6.6 \times 10^{-6}}{(0.20)^2} = x$$

$$1.65 \times 10^{-4} = x$$

$$[\text{Pb}^{2+}] = 1.7 \times 10^{-4} \text{ mol/L}$$

From dissociation equation, ratio of $\text{PbBr}_2(s)$ to $\text{Pb}^{2+}(aq)$ is 1:1

$$[\text{PbBr}_2(s)] = 1.7 \times 10^{-4} \text{ mol/L}$$

The molar solubility is $1.7 \times 10^{-4} \text{ mol/L}$